

Harshil Gorasiya *M.Sc. Student, AI/ML Engineering*

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harshil-0011.github.io/My-Portfolio

Profile

M.Sc. student in Software Engineering and Management (Hochschule Heilbronn) focused on applied AI/ML engineering. Built and evaluated local RAG systems, a multi-provider LLM coding agent, and computer-vision models; currently working on designing and training a language model from scratch. Seeking a working student (Werkstudent) role or Master's thesis in AI/ML engineering.

Technical Skills

Programming: Python | C++ (multithreaded, POSIX threads) | SQL | Bash

LLM Development: Transformer architecture | tokenizer training (BPE) | pre-training loop | mixed-precision or multi-GPU training | fine-tuning (LoRA/QLoRA) | training-loss/perplexity tracking

LLMs, RAG & Agents: RAG | Knowledge graphs | FAISS | Ollama | ReAct agents and tool use | Anthropic API | MCP | Quantization (GGUF/AWQ) | Multimodal inference (LLaVA) | OCR (Tesseract | LangChain | SentenceTransformers

ML & Computer Vision: TensorFlow/Keras | YOLOv8 | CNNs with residual blocks | PyTorch | Scikit-learn | Hugging Face Transformers | Real-ESRGAN

Engineering & MLOps: Git | Docker | Linux | REST APIs | FastAPI | MLflow

Evaluation: Validation accuracy | latency (incl. query latency vs. corpus size) | throughput (FPS) | false-positive analysis | PSNR | SSIM

Additional tools: Azure | AWS | GCP | PostgreSQL | MongoDB | SQLite | Power BI | Tableau

Education

M.Sc. Software Engineering and Management, **Hochschule Heilbronn** *2025 – Present*
Heilbronn, Germany

B.Tech. Information Technology, **A. D. Patel Institute of Technology** *2020 – 2024*
Gujarat, India

Technical Projects

Ardan-CLI, **Autonomous Multi-Provider Coding Agent**

- Built a coding agent (~4,000 lines of Python) that runs a ReAct-style loop: plan, execute, observe, auto-correct, review.
- Implemented latency-based failover across six LLM providers (Claude, GPT-4.1, Gemini, Groq, Mistral, Ollama).
- Added parallel tool execution for Git, Docker, C++ compilation, and test runners to automate software-development workflows.
- Added AES-256 encrypted key storage, multimodal image input, and session undo/diff.

Anvitha RAG, **Knowledge Graph Retrieval System**

- Built a fully local RAG pipeline (FAISS, Ollama) that structures documents into a five-level knowledge graph, from document down to sentence; indexed 75 pages (225 KB) in 19 seconds.
- Combined semantic (FAISS), BM25, and graph retrieval, with source tracking and citations in answers.
- Measured an average query latency of 4 seconds over 10 queries and five corpus sizes (9 KB to 226 KB), with no significant degradation as the corpus grew.
- Developed as a replacement for a Microsoft GraphRAG prototype.

- Added query rewriting, adaptive retrieval depth, metadata-aware ranking, and response self-review.

Local Perplex, **Offline Multimodal Research Engine**
- Implemented a native C++ multithreaded ranker (producer-consumer queue, POSIX threads) with under 10 ms ranking latency across 10,000+ documents.
- Built a retrieval pipeline for multimodal research with 3–8 second processing time.
- Runs fully offline: local OCR (Tesseract), vision inference with quantized LLaVA, and multi-format RAG storage for PDF, Word, Excel, CSV, and Markdown.

AI Video Upscaling, **Windows Desktop Application**
- Developed a Windows desktop application using Real-ESRGAN super-resolution with dynamic model selection and GPU monitoring.
- Evaluated video upscaling performance using PSNR and SSIM on a standardized 30-clip benchmark
- AI Video Upscaling
- Built an AI video upscaling pipeline with CPU/GPU selection, evaluated with PSNR and SSIM.

LLM from scratch

Language Model from Scratch (in progress)
- Designing a language model architecture and training it from scratch (work in progress).

Professional Experience

AI & Web Solutions Developer, **Aani Brother Infotech** *05/2024 – 02/2025*
Surat
- Built web applications by translating business requirements into user interfaces and application workflows.
- Contributed to web projects involving AI-enabled solutions, incorporating functional requirements into the user experience.
- Conducted user research, wireframing, and prototyping, and iterated on designs to improve usability and consistency.

Deep Learning Intern, **PD Avenue** *12/2023 – 04/2024*
Surat
- Developed a TensorFlow/Keras residual-block CNN for age and gender classification on a 22,000-image dataset; raised validation accuracy from 71% to 79% through architecture optimization and class-balancing augmentation.
- Reduced training time on V100 GPUs by 20% with cosine learning-rate scheduling and larger batch sizes, while maintaining accuracy.
- Identified and resolved a data-leakage problem, improving experiment reliability across 10+ structured training runs.

Machine Learning Intern, **Dotcom IoT LLP** *05/2023 – 06/2023*
Surat
- Implemented a YOLOv8n object-detection pipeline on a CPU-only edge device (Intel Core i7), processing 720p video at 28–32 FPS.
- Achieved 2.4× higher throughput than a YOLOv5s baseline; curated and annotated 3,000+ images across five vehicle classes.
- Tuned IoU and NMS parameters, reducing false positives in low-light conditions by 18% on a dedicated night-test dataset.

Certifications

Anthropic: Claude with the Anthropic API: 2026
Anthropic: Model Context Protocol – Advanced Topics: 2026
Databricks: Fundamentals of Data Science: 2025
IBM: Machine Learning with Python: 2025

Languages

English: Fluent (full professional proficiency)
German: Good (professional working proficiency)